DEFINING AI-AUGMENTED ACADEMIC PROJECTS

When AI Helps and When It Doesn't

AI Days 2026 · Abbie Petulante, Postdoctoral Research Fellow

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So You Want to Use AI

1 Try it

You have a task

You ask Claude, ChatGPT, Gemini, etc.

It produces something quickly and with little effort on your part

2 Then...

First pass looks great, seems to tick the boxes

You're done?

3 Or...

Oh no, that's not what you wanted at all, it needs a lot of fixes

Give up?

Neither outcome is good.

A Real Example

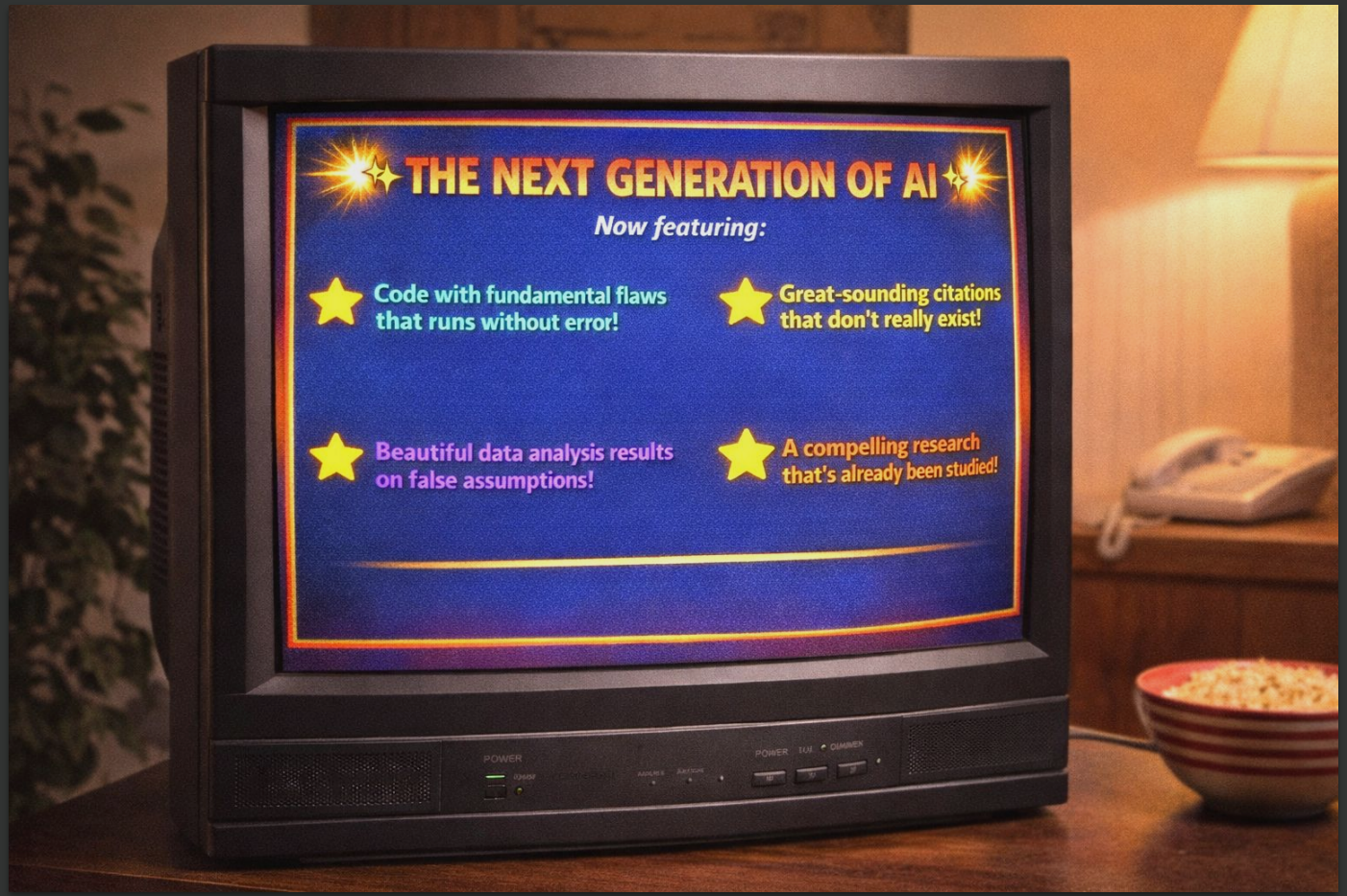
The task: Generate comprehension questions for children's books

The teacher is looking at all her students on the computer screen. Can you tell me about a time when you learned something new that made you feel excited?

What do you think the teacher and the kids are learning together on the computer?

The reality: ~50% of questions are poor. Extensive prompt engineering with state-of-the-art models: ~80% quality - below our needs

It can feel like this at times.

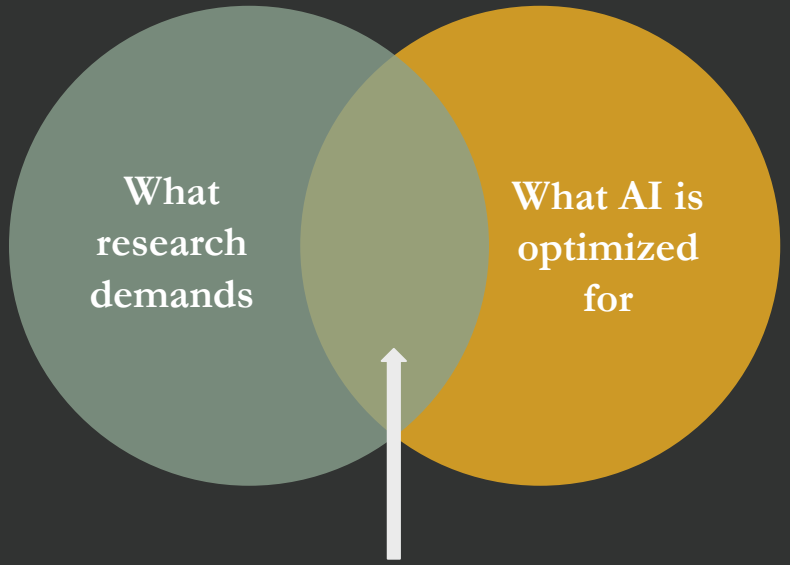


AI is genuinely incredible.

It can process thousands of papers in minutes. It can write functional code from a description. It can find patterns humans miss. It can accelerate every stage of the research lifecycle - *when used well.*

Research is where the details matter most

- Correctness
- Edge case coverage
- Principled methodology
- Reproducibility
- Subtle errors can be catastrophic



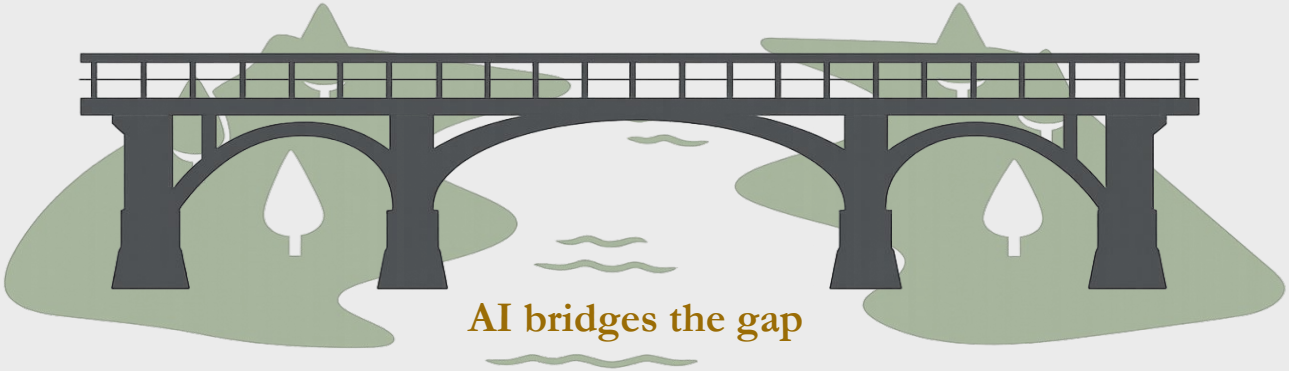
- Plausibility
- Fluency
- Speed
- Breadth
- Pattern-matching and recognition

There is a lot in the middle!

How do we know what goes there?

Bridging the Gap, Not Plane-ing it

You have expertise. You want to move to a problem space where that expertise is critically informative, but the task is just out of reach.



You have expertise. You want to try something complex and totally new - your expertise doesn't transfer.

AI is most powerful when it closes the production gap for someone who already has the evaluation expertise.

AI is most dangerous when it closes the production gap for someone who doesn't.

AI Failures are Different

The signal that something is wrong is absent

Bad AI code runs perfectly, and bad AI writing reads beautifully. There's no error message, no warning flag. You have to go looking for the problem.

AI inverts the normal cost structure

Usually the hard part is doing the work and evaluating is easier. AI flips - producing is trivial, evaluating is very hard.

AI's self-explanations are subject to the same trap

It’s difficult, when decisions are opaque or confusing or something seems wrong, to *really* understand an AI’s thought process

Four Modes of AI in Research

AI as Information Processor

Feeding data through an LLM,
using structured output

AI as Writing/ Thinking Partner

Drafting, editing, literature synthesis, brainstorming

AI as Your Programmer

You design the experiment,
AI writes the code

AI as the Thing You're Building

Training/fine-tuning a model
as part of your research

Five Pressure Points

The fundamental ways AI goes wrong in research, how this shows up in the modes, and what you can do about each one

1

The Invisible Decision Problem

AI makes choices you didn't ask for and doesn't tell you

Programming

Picks a "simpler" algorithm, chooses defaults, handles edge cases its own way

Building

Training shortcuts, architecture decisions, data preprocessing choices

Processing

How it interprets ambiguous categories, what counts as a "match"

Writing

What it emphasizes, what framing it chooses, what it leaves out

1

Make the decisions visible

You can't control what you can't see

- → Ask AI to state its assumptions before executing. “Before we do this task, explain how you plan to tackle it. What choices are you making?”
- → Write specs that constrain choices
- → Treat the first pass as a draft of decisions. Start small, ask for pieces of tasks a lot of invisible decisions one at a time

2

The Plausibility Trap

The output looks right, it runs, it sounds authoritative. It can still be wrong.

Programming

Code executes without errors but contains a methodological flaw

Building

Model outputs pass a vibe check but fail systematic evaluation

Processing

Each individual classification seems reasonable — until you check

Writing

The paragraph reads great; the work it's referencing doesn't exist

2

Don’t trust the vibes

*Things need to **be** right, not just seem right*

- Start with a test case you already know the answer to
- Check the edges, not the middle.
- Verify claims independently. Let someone take a second look!
  Check citations. Trace the logic on a known input. Sanity-check the numbers.

The Evaluation Gap

Checking AI's work requires the expertise AI was supposed to save you from needing

Programming

You need to read and understand the code to catch minor but critical mistakes

Building

You need domain-specific metrics, not just loss curves

Processing

You need sampling and validation strategy that measure correctness

Writing

You need to fact-check - which means doing the literature review anyway

3

Choose problems that leverage your expertise

The difficulty of the evaluation depends on the problem

- Know what evaluation means for your specific problem. Will it be obvious when you look at it? Or when you run extensive validation?
- Try to scope ahead of time. How much can you anticipate right now? How much do you need to "wait and see"?
- Ask yourself if you could learn it if you need to.

The evaluation gap isn't fixed - it's a spectrum. Sometimes the answer really is "this is easy to check, so AI is a great fit."

The Aggregation Effect

One output looks fine. One thousand reveal systematic issues.

Programming

One subtle bug doesn't cause issues in your test pipeline. It does in production.

Building

Consistent bias across all model outputs compounds at scale

Processing

Nuanced categories get flattened across a whole corpus

Writing

Everything you write sounds the same. You've made this point 10 times now.

4

Check the distribution, not the instance

Don’t trust one output when it needs to work more than once

- Sample strategically, not randomly
- Look at statistics across a large variety of cases.
- Know what biases and errors are acceptable. What happens if your outputs are biased? Lack diversity? Contain errors?

The Cost Illusion

Sometimes management is more work than doing it yourself

Programming

Debugging complex AI code can take longer than writing it yourself

Building

A fine-tuning project that saves 10 hrs might cost 100 hrs to get right

Processing

Building a reliable validation pipeline is a research project itself

Writing

Editing AI prose into your voice can take longer than drafting

5

Prototype before you commit

Find out what it costs before you've invested too much time

- **Do a small pilot first.** Include the full evaluation, debugging, and iteration pipeline. Is it holding up?
- **Compare honestly against the old way.** Sometimes it's really just faster to do things yourself. That's ok.
- **Budget more time than you think**

So when do I use AI for this?

A framework for deciding whether and how to use AI in your research. Six questions to ask before you start

Q1 What exactly is the task?

Problem decomposition

I can write a precise spec with clear inputs and outputs	→ Proceed
I have a general idea but the details are fuzzy	→ Define it further before starting AI
I'd know it if I saw it, but can't specify it	→ Use caution, just explore until you have details

Q2 What's your baseline?

What does 'good' look like without AI?

I know what quality looks like and how long it takes, now	→ Proceed, and measure improvement
I've never done this task before, but I understand the domain	→ Establish a small manual baseline first
I have no reference point for what 'good' looks like	→ You can't evaluate if AI is doing quality work.

Q3 What's your error tolerance?

What are the stakes when something is wrong?

Errors are easily caught in later stages, by reviewers, or are relatively harmless	→ Proceed, build in those review steps
Some errors are acceptable, but systematic errors aren't	→ Decide systematic tests, check aggregation effects
A single error could be catastrophic, and subtle enough to miss	→ Be extremely rigorous with evaluation or no go

Q4 Can you evaluate the output?

The single most important question

I'd know immediately if something was wrong, and likely *what* was wrong	→ Great fit, AI closes the production gap
It will take some to formalize, but I know how	→ Design validation and decision frameworks
I don't know of a way to check, and don't know what could go wrong	→ The gap is too big. More work needed first.

Q5 What are the hidden decisions?

What will AI decide that you didn't specify?

I can anticipate the major choices, constrain them, and identify unanticipated ones	→ Write your specs down, review extra decisions
I can't anticipate what choices it will make, but would know bad ones	→ Use the 'list assumptions' tactic, review carefully
I don't know what could be assumed, and I wouldn't know a decision was bad	→ Back to Q1, reformulate. What's missing?

Q6 What's the real cost?

Look at all of your answers. What is AI buying you?

I answered mostly or all green	→ There is clear opportunity for AI to accelerate
There were several yellows, but I have a plan for them	→ Start with a small pilot. Is effort as expected?
I had multiple red flags	→ AI might make life harder

What Now?

I'm good to use AI!

or

No AI for me!?

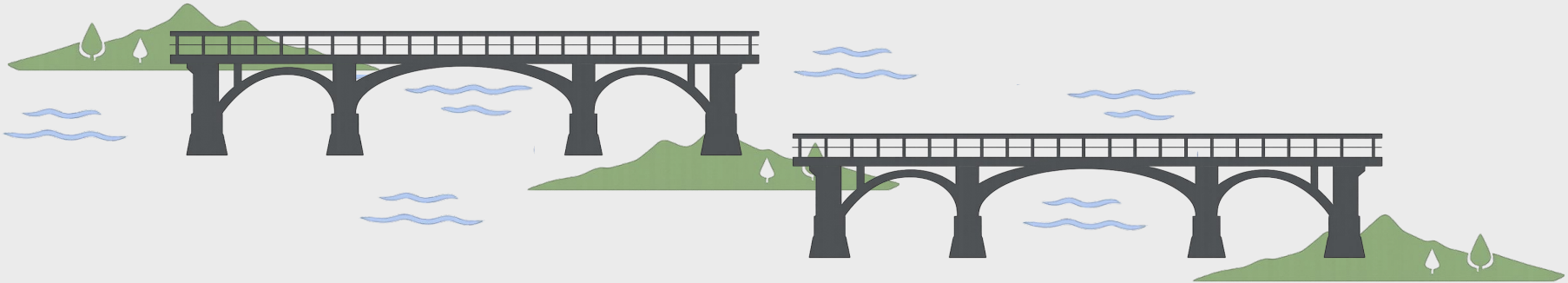
Of course not!

If a portion of your project isn't a fit, it doesn't mean the whole project isn't a fit

- Decompose into smaller parts
- Run the whole framework again on just that part

Q7 What's a piece AI *can* help with?

Enough bridges cross an ocean.



Final Thoughts

When AI Tends to Disappoint

- You don’t know what “done right” is
- Checking the output means just doing it
- Your needs aren’t concrete or well-defined
- You can’t carefully evaluate
- Managing > the work it replaces

When AI Tends to Work Well

- High volume, reproduced reliably
- Clear, evaluable success criteria
- Human review is easily built in
- Experts are augmented, not replaced
- Errors are low/moderate stakes

The Middle Ground

But, the reality is many projects live here:

AI could work for this, but it will take investment in evaluation, iteration, and human oversight - sometimes a lot

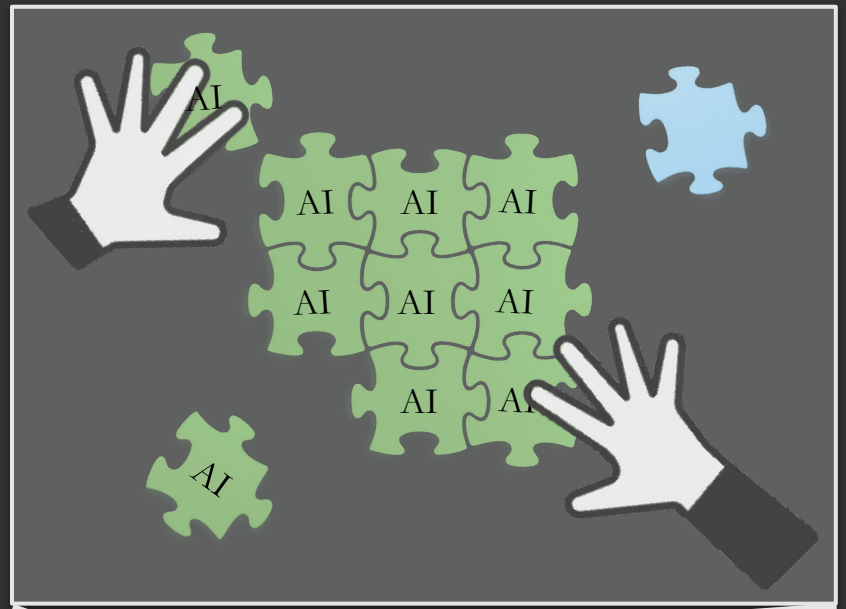
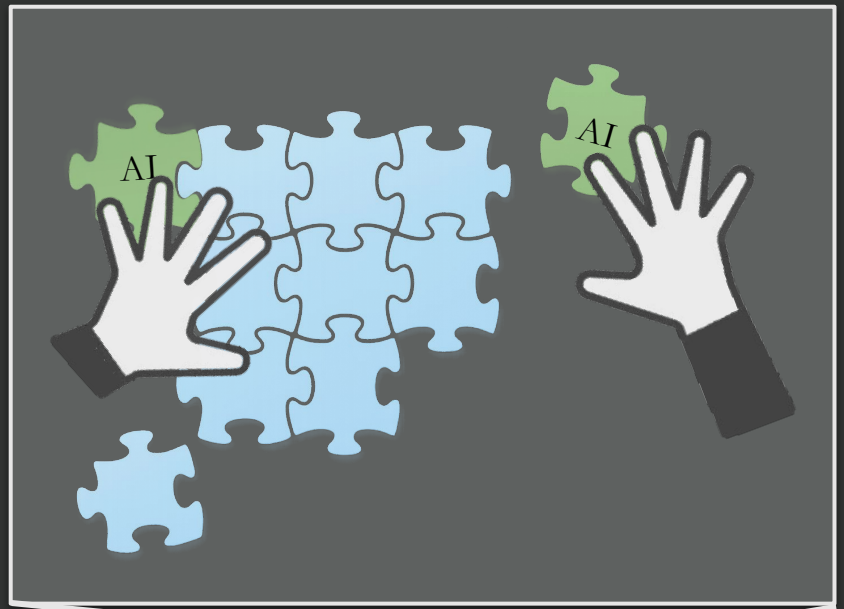
The honest question isn't "can AI do this?", but "is that investment in making AI do this **well** worth it, compared to alternatives?"

What is the piece that *is* worth it?

Evaluation is the name of the game

Build it in since day one. Evaluate more things and more often than you might normally think to

- Define the metrics and scope before you start
- Establish baselines
- Check the implementation, not just the output
- Document everything
- Plan for iteration

You are still the puzzle solver

AI can be some of the pieces